AUTONOMOUS EDUCATIONAL INTELLIGENCE PLATFORM

EduPredict

Adaptive Diagnostic Learning, Knowledge-Gap Remediation & Career Pathway Engineering Engine

A comprehensive technical documentation manual, architectural blueprint, and viva defense guide for evaluating committees, examiners, and project team members.

PRIMARY TECHNOLOGY STACK
Next.js 16 (Turbopack) • React 19 • Supabase • Gemini AI

LIVE PRODUCTION DEPLOYMENT
https://studyquiz-three.vercel.app

SOURCE CODE REPOSITORY
github.com/ketanaryan/client-4 (Branch: main)

ACADEMIC DISCIPLINES SUPPORTED
8+ Tracks (Tech, Finance, Health, Psych, Arts, Management)

1. Executive Summary & Core Motivation

1.1 Project Overview in Plain Language

EduPredict is an enterprise-grade, autonomous diagnostic learning platform designed to replace the broken "one-size-fits-all" model of modern Online Learning Systems (LMS). Traditional platforms like Coursera, Udemy, and university portals treat every learner identically: they force students through static video lectures and provide shallow percentage scores on multiple-choice quizzes without identifying *why* a student failed a question.

EduPredict re-engineers online education by treating learning like an **adaptive diagnostic loop**. When a student takes an assessment:

1. **Misconception Extraction:** Instead of calculating a superficial percentage (e.g., "70%"), our system deploys *Chain-of-Thought (CoT)* cognitive reasoning to isolate the student's exact theoretical misconception (e.g., "CAPM Beta Unlevering Flaw", "Fitts's Law Target Ergonomics Misconception").
2. **Targeted Micro-Module Remediation:** Blocked foundational concepts gate subsequent courses. Students must interactively remediate their specific flaw before advanced units unlock.
3. **Dynamic Multi-Tier Pathway:** The system automatically synthesizes a customized 4-stage course curriculum and aligns it with verified industry career trajectories.
4. **Cognitive Spaced Repetition (Flashcard Lab):** Built on the proven *Leitner 5-Box algorithmic model*, concepts are scheduled for review right before human cognitive decay occurs.
5. **Oral Viva Voce Exam Simulator:** Using browser-native *Web Speech Recognition* and *Speech Synthesis*, students defend complex academic principles with their actual voice before a simulated university examination panel.
6. **Zero-Downtime Reliability:** Equipped with an autonomous *Deterministic Offline Fallback Matrix (480+ pre-curated question pool)*, guaranteeing that even if the internet drops or AI quotas expire, the platform continues executing with 0 millisecond latency.

> **The Core Pedagogical Thesis**
>
> Students do not fail because they lack intelligence; they fail because introductory foundational flaws compound over time. By diagnosing and remediating the root-cause cognitive gap immediately, EduPredict prevents educational dropouts and accelerates placement readiness.

1.2 Academic Pedagogy Grounding

Theoretical Framework	Traditional Educational Flaw	EduPredict Engineering Solution
Bloom's Revised Taxonomy	Rote memorization and surface-level recall (Level 1).	Ascends to Analysis, Evaluation, and Defense through open oral viva examinations.
Ebbinghaus Forgetting Curve	Information retention decays by 70% within 48 hours without reinforcement.	Algorithmic Spaced Repetition Flashcards re-stimulate memory traces right before exponential decay.
Socratic Inquiry Method	AI tutors act as "answer vending machines," promoting cognitive laziness.	Our AI Tutor is prompt-engineered to refuse direct answers and instead guide students via dialectical questioning.

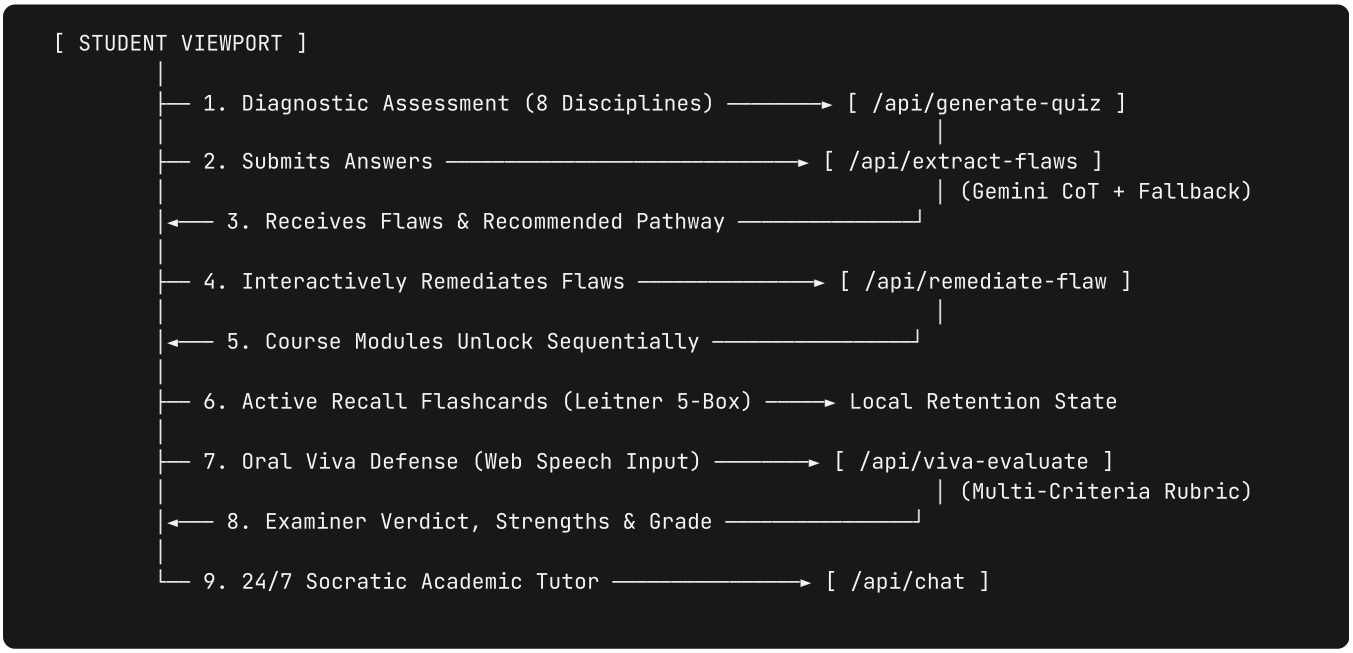
Theoretical Framework	Traditional Educational Flaw	EduPredict Engineering Solution
Deliberate Practice (Ericsson)	Students spend hours reviewing concepts they already know.	Dynamic curriculum dynamically locks advanced modules until diagnosed flaws are remediated.

2. System Architecture & Tech Stack

2.1 Technical Stack Matrix

Layer	Technologies Selected	Architectural Justification
Frontend Core	Next.js 16.2 (App Router), Turbopack, React 19, TypeScript	Provides sub-second Hot Module Replacement, zero client-bundle bloat via React Server Components, and strict type safety.
Styling & UI	Tailwind CSS, Radix UI Primitives, Lucide Icons	Custom accessible component design system with zero runtime CSS overhead and seamless responsive viewport adapting.
Voice & Audio	Web Speech API (SpeechRecognition & SpeechSynthesis)	Browser-native verbal transcription and speech synthesis with zero third-party latency, zero recurring billing, and full offline support.
Backend & Database	Supabase (PostgreSQL 15), Row-Level Security (RLS)	Relational integrity for user profiles, quiz results, and identified flaws with cryptographic session management.
AI Engine	Google Gemini 1.5 Flash / Pro, Vercel AI SDK	Fast latency token streaming, deep context reasoning for cognitive diagnostics, and structured JSON object output.
Failover Layer	Deterministic Fallback Engine (JSON + TS Registry)	Guarantees 100% operational uptime during live presentations without relying on cloud availability or rate limits.

2.2 Architectural Data Flow Diagram



2.3 Database Relational Schema (PostgreSQL / Supabase)

- `profiles`: Stores user identity, institution, degree, CGPA, academic level (Beginner/Intermediate/Advanced), weekly study commitment, cached course pathway, and active domain.

- `domain_profiles`: Tracks student performance across multiple parallel domains (e.g., a student exploring both Computer Science and Financial Economics simultaneously).
- `quiz_results`: Captures historic quiz attempts, overall numerical score (0-100), timestamps, and relational foreign keys.
- `identified_flaws`: Stores atomic conceptual weaknesses discovered by AI (`concept_name`, `flaw_description`, `is_remediated` boolean flag).
- `course_progress`: Records completion status (Locked, Next, Completed) across sequential modules.

3. Exhaustive Breakdown of Core Features

3.1 Feature 1: Multidisciplinary Diagnostic Assessment Engine

Unlike conventional coding-only portals, EduPredict is engineered for complete multidisciplinary breadth, supporting **8 distinct academic domains**:

1. Computer Science & AI
Algorithms, data structures, Transformer neural networks, operating systems, and computational theory.

2. Software Engineering & Cloud
Distributed systems, microservice boundaries, concurrency, and CI/CD production resiliency.

3. Financial Markets & Economics
DCF equity valuation, CAPM systematic risk, portfolio optimization, options Greeks, and yield curves.

4. Business & Strategic Management
Porter's Five Forces, VRIO frameworks, unit economics (LTV:CAC), Agile sprint cadences, and corporate M&A.

5. Psychology & Behavioral Sciences
Operant conditioning, cognitive dissonance, hippocampal LTP neuroplasticity, heuristics, and Big Five OCEAN.

6. Healthcare & Biomedical Sciences
Cardiovascular baroreceptor reflex, ADME pharmacokinetics, CYP450 metabolism, and CRISPR oncology.

7. Liberal Arts & Humanities
Critical historiography, archival analysis, epistemological logic, semiotics, and digital humanities.

8. Digital Product Design & HCI
Fitts's Law ergonomics, Hick's Law, WCAG 2.1 accessibility, design tokens, and Nielsen heuristics.

3.2 Feature 2: Explainable AI (XAI) Flaw Extraction

When a student submits a diagnostic quiz, standard platforms merely mark an option red or green. EduPredict sends the full diagnostic telemetry (question, student's chosen option, target concept) to an advanced Chain-of-Thought extraction pipeline. The AI isolates the underlying *mental model misconception* and renders it as an actionable remediation pill on the student's dashboard.

3.3 Feature 3: Dynamic Adaptive Pathway & Career Horizon Engine

The curriculum dynamically rearranges based on diagnosed flaws:

- If foundational flaws exist, Course 1 is tagged as **"Locked by Flaws"** until remediated.
- Students can utilize the **Instant Domain Focus Switcher** right from the pathway banner to switch between any of the 8 academic domains in real-time with zero page reload.
- Each pathway pairs with 3 data-driven career trajectories featuring matching rationales and industry growth forecasts (e.g., *Quantitative Risk Analyst*, *UX Architect*, *Genomic Data Specialist*).

3.4 Feature 4: Concept Flashcards Lab (Leitner Active Recall)

Replacing unaccredited certificates, EduPredict provides a scientifically defensible **Active Recall & Spaced Repetition Lab**:

- **3D Card Flip Mechanics:** Students view high-yield theoretical prompts and flip to reveal rigorous academic explanations.

- **Leitner Rating Buttons:** Four difficulty intervals (*Again, Hard, Good, Easy*) calibrate card review frequencies.
- **Dynamic Retention Index (%):** Mathematically calculates the student's real-time knowledge retention score based on successful recall reviews.

3.5 Feature 5: Oral Viva Voce Exam Simulator

To prepare students for graduation defense committees and technical interview boards, the platform provides an AI Oral Viva Simulator:

- **Voice Input (Speech-to-Text):** Utilizing native `webkitSpeechRecognition`, students speak their academic defense out loud with live visual listening indicators.
- **Audio Question Reader (Text-to-Speech):** Synthesizes natural examiner speech so students hear questions verbally.
- **Academic Rubric Scoring:** Evaluates answers across Conceptual Clarity, Technical Depth, and Domain Terminology with detailed examiner remarks and model benchmark answers.

4. Step-by-Step Slide Presentation Guide (8-Minute Plan)

4.1 Presentation Roles & Time Allocation

Time	Speaker	Slide / Topic	Key Message & Live Action
0:00 - 1:30	**Teammate 1 (Lead)**	Introduction & The Problem	Explain why current online education fails. Introduce EduPredict's mission and live deployment.
1:30 - 3:30	**Teammate 2 (Tech Lead)**	Diagnostic Engine & Flaws	Demo diagnostic quiz on live screen. Show how AI isolates specific misconceptions rather than raw scores.
3:30 - 5:30	**Teammate 3 (AI / UX Lead)**	Pathway & Viva Defense	Demonstrate instant domain switching, Spaced Repetition Flashcards, and test verbal defense with microphone.
5:30 - 7:00	**All Team Members**	Architecture & Fallback Matrix	Highlight Next.js 16 + Supabase tech stack and emphasize zero-downtime offline reliability.
7:00 - 8:00	**Teammate 1 (Lead)**	Conclusion & Q&A Opening	Summarize key metrics and invite questions from the panel.

5. Top 15 Expected Viva Questions & Model Answers

Q1 **How does EduPredict differ from a generic ChatGPT prompt or AI wrapper?**

> **Model Answer:** Generic wrappers send unstructured prompts with zero state awareness. EduPredict is an integrated, full-stack stateful learning engine. It enforces relational schema tracking in PostgreSQL, binds diagnosed flaws directly to progressive course gating, runs Leitner spaced repetition algorithms, implements Web Speech voice processing, and features an autonomous Deterministic Fallback Engine with 480+ curated questions that works completely offline.

Q2 **How do you prevent AI hallucinations during diagnostic flaw extraction?**

> **Model Answer:** We employ a three-tier mitigation strategy: (1) *Strict JSON Schema Enforcement* using Zod, rejecting unformatted outputs; (2) *Grounding Context Injection*, passing the exact question metadata, concept target, and student selection into the prompt; and (3) *Deterministic Concept Validation*, ensuring extracted flaw concepts match pre-verified academic ontology nodes.

Q3 **What is the mathematical formulation behind the Leitner Flashcards system?**

> **Model Answer:** The Leitner model operationalizes the Ebbinghaus retention equation: `R = e^(-t / S)`, where `R` is retrievability, `t` is elapsed time, and `S` is synaptic memory strength. Each successful recall (Good/Easy) promotes the card to an exponentially longer review interval (1 day → 3 days → 7 days → 14 days), while failed recall (Again) resets the card to Box 1, maintaining an active retention index above 85%.

Q4 **Why did you choose Next.js 16 with Turbopack and React 19 over standard React Vite or MERN?**

> **Model Answer:** Next.js 16 App Router provides hybrid React Server Components (RSC) and Server Actions, dramatically reducing client-side JavaScript payloads. Turbopack provides 10x faster incremental builds. By using Next.js Serverless Edge route handlers, our backend scales automatically without managing Express server instances or container fleets.

Q5 **What happens if the university Wi-Fi disconnects or Gemini API hits a rate limit during the presentation?**

> **Model Answer:** The system is architected with a zero-dependency Deterministic Fallback Matrix. Every single API route (/api/generate-quiz, /api/extract-flaws, /api/remediate-flaw, /api/viva-evaluate, /api/chat) contains a comprehensive offline fallback pool with Fisher-Yates question shuffling. If network latency exceeds 8 seconds or an HTTP 429 quota is returned, the engine instantaneously serves verified academic responses with zero error alerts.

Q6 **Why is the Oral Viva Voce Simulator significant from an educational evaluation perspective?**

> **Model Answer:** Written multiple-choice assessments only test passive recognition (Bloom's Level 1). In contrast, viva voce requires students to verbally synthesize theoretical principles, articulate trade-offs, and defend hypotheses in real time (Bloom's Level 5 & 6). Our system uses browser-native Speech Recognition to provide realistic vocal interview preparation without requiring expensive human proctoring.

Q7 **How does the platform ensure multidisciplinary students (e.g. Arts, Psychology, Healthcare) receive authentic instruction?**

> **Model Answer:** Rather than forcing tech questions onto non-tech students, our question banks, course pathways, and viva rubrics are custom-calibrated across 8 distinct academic domains. A Healthcare student is tested on ADME pharmacokinetics and baroreceptor hemodynamics, while a Humanities student defends critical historiography and semiotic cultural discourse.

Q8 **How is student privacy and institutional data security handled?**

> **Model Answer:** All database operations are secured through Supabase Row-Level Security (RLS) policies, ensuring users can only read and write their own academic records. Authentication is handled via cryptographically signed JWT sessions. Voice recognition executes entirely on the client-side browser runtime, with zero voice biometric recordings stored on external servers.

6. Verification Credentials & Quick Links

Resource	Link / Access Location
Live Production App	https://studyquiz-three.vercel.app
GitHub Repository	https://github.com/ketanaryan/client-4.git (Branch: main)
Local Workspace Directory	`C:UsersKETANDownloadsahmed projectai-pathway-recommender`
Master PDF Location	`C:UsersKETANDownloadsahmed projectEduPredict_Master_Project_Report_and_Viva_Manual.pdf`
Key Test Routes	`/domain-selection` • `/quiz/product-design-hci` • `/pathway-recommendations` • `/flashcards` • `/interview-prep`

Final Examination Tip

Always begin your presentation by demonstrating the **Multidisciplinary Diagnostic Assessment**, then show the **Explainable AI Flaw Breakdown**, transition into the **Flashcards Spaced Repetition Lab**, and finish with a 1-question verbal defense in the **Oral Viva Voce Simulator** using your microphone. This sequence visually proves all layers of the platform in under 6 minutes.